G Case study: Hallucinations in LLM's answer and how our LLM Eval pipeline detects them

Basic record (Row 369, *Sodium Correction for Hyperglycemia* calculator)

Patient 57-year-old **male**; $\mathrm{Na}_{\mathrm{meas}} = 127$ mmol/L (127 mEq/L); glucose $=$ 527 mg/dL

Clinical note

Diabetic foot with massive hyperglycaemia and haemodynamic instability (full narrative in dataset)

Question "What is the corrected sodium concentration (mEq/L) using the Hillier 1999 equation at admission?"

Gold-standard reasoning

Hillier's formula (**?**):

$$\mathrm{Na}_{\mathrm{corr}} \;=\; \mathrm{Na}_{\mathrm{meas}} \;+\; 0.024\,(\text{glucose} - 100).$$

$$127 + 0.024\,(527 - 100) = \mathbf{137.248}\ \text{mEq/L}.$$

Baseline tolerance (dataset,±6.8624): $[130.39,\ 144.11]$ mEq/L.

LLM original answer (excerpt)

"Corrected sodium (mEq/L) = 127 + 0.016×527 $= 127 + 8.432 = 135.432$ *mEq/L."*

Why the baseline benchmark says "Correct"

The Medcalc benchmark inspects only whether the final number lies within the broad interval above. Because $135.432 \in [130.39, 144.11]$, the response is labelled **"Correct"**, even though the equation is mis-specified.